

Windowed DMD as a Microtexture Descriptor for Finger Vein Counter-spoofing in Biometrics

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Abstract—Recent studies have shown that it is possible to attack a finger vein (FV) based biometric system using printed materials. In this study, we propose a novel method to detect spoofing of static finger vein images using Windowed Dynamic mode decomposition (W-DMD). This is an atemporal variant of the recently proposed Dynamic Mode Decomposition for image sequences. The proposed method achieves better results when compared to established methods such as local binary patterns (LBP), discrete wavelet transforms (DWT), histogram of gradients (HoG), and filter methods such as range-filters, standard deviation filters (STD) and entropy filters, when using SVM with a minimum intersection kernel. The overall pipeline which consists of W-DMD and SVM, proves to be efficient, and convenient to use, given the absence of additional parameter tuning requirements. The effectiveness of our methodology is demonstrated using FV-Spoofing-Attack database which is publicly available. Our test results show that W-DMD can successfully detect printed finger vein images because they contain micro-level artefacts that not only differ in quality but also in light reflection properties compared to valid/live finger vein images.

I. INTRODUCTION

Biometric techniques for person authentication have significant advantages over keys, passwords, and PIN numbers because these conventional authentication mechanisms can be stolen, lost, or forgotten. Hence we have seen substantial progress in research and deployment of different biometrics for person authentication such as face, finger print, iris, finger vein (FV), signature, and other modalities. However, these biometric traits can also be spoofed for malicious reasons. Finger prints and palm prints can be replicated using everyday materials such as silicon gel, gums, etc [1]. Facial authentication systems can be spoofed using printed, replayed media and other commonly available materials [2]. Many successful spoofing attempts reported in the literature using commercial systems, e.g., [3], [4]. Countermeasures are thus being developed to neutralise attacks [5]. Using the finger vein biometric trait as an authentication method has gained popularity in recent years, particularly within the financial sector. Unlike finger prints, finger vein patterns exist inside the human body, thus eliminating tampering and providing an ideal identification method without being intrusive. It is widely used in Japan, China, Poland and Turkey [6] and recently banks in the UK have also planned to introduce this technology¹. However, a recent successful spoofing attempt on finger vein biometric was reported by IDIAP [7]. Although the attack was carried out under the users' cooperation, this nevertheless shows the vulnerability of the system.

A. Existing countermeasures

IDIAP initiated an open competition for FV-spoofing counter-measures. Their article [6] reported 4 different approaches from the competition, which are briefly described here.

IDIAP proposed a texture-based algorithm exploiting the frequency domain of the images. The results from their algorithm were the baseline for the competition. GUC team used a texture-based approach extracting binarized statistical image features (BSIF) [8], which represent each pixel as a binary code obtained by computing its response to a filter [9] learnt using the statistical properties of natural images [10]. B-lab team utilised monogenic scale space (MSS) [11] based global descriptors. This method extracts the local energy and local orientation at a coarse level, allowing location of the difference between spoofed and non-spoofed vein blocks. The winning team, GRIPPRIAMUS used local descriptors such as 3-LBP_{8,1}^{u2} [12] for cropped images and Local Phase Quantization (LPQ) [13] and Weber Local Descriptor (WLD) [14] on full images. The results from the competition show that texture based methods are one of the best counter-measure strategies for detecting the finger vein spoofing attacks especially on printed media.

Despite the difference in the nature between finger vein counter spoofing and finger vein recognition, both tasks typically rely on texture or feature extraction with respect to a single finger vein image acquired using an infra red or near infra red camera. Our conjecture is that our proposed W-DMD method will thus also differentiate spoofed and non-spoofed finger vein biometrics access using a single image.

Our proposed method, W-DMD is a variant of the DMD method originally developed in the area of computational fluid dynamics (CFD) [15]. Specifically, DMD, is a mathematical method developed to extract the relevant modes from empirical data generated by non-linear complex fluid flows [16], [15], [17], [18]. DMD was introduced to the field of biometrics-spoofing by Tirunagari *et al.* [2] to distinguish a live access from printed media or replayed videos by exploiting the information dynamics of the video content, such as blinking eyes, moving lips, and facial dynamics in an entirely data-driven approach. In general, DMD is used to analyse a set of image sequences; in this paper, however, we present an approach, W-DMD, exclusively for analysing an image in order to detect spoofed finger vein images. We also aim to find answers for the following questions:

- 1) If DMD can capture temporal information then would W-DMD capture texture information?
- 2) If DMD can capture principle movements then would W-DMD capture texture gradients?

¹<http://www.bbc.co.uk/news/business-29062901>

- 3) What would be the effect of texture gradients on classification performance ?
- 4) How effective is the W-DMD compared to plethora of existing descriptors ?

B. Preliminary analysis: Window-DMD

Having examined the images of valid versus spoofed finger vein image samples, we observe that both types of images differ in terms of image quality and light reflection. For example, a valid finger vein image and a printed one reflect light in different ways because a vein is a complex non rigid 3D object whereas a photograph can be seen as a planar rigid 2D object. This may cause different reflections and shading. The surface properties of valid finger vein compared to the printed ones are also different. We therefore propose to solve the finger vein spoof detection problem by analysing the finger vein for capturing the aforementioned information at micro-texture level.

1) *Experimental analysis: Visual examination:* The effectiveness of the proposed W-DMD is demonstrated for cropped finger vein images in Figure 1 and for full images in Figure 2. The top row in these figures show the raw images for the valid access and spoof, whereas the middle and bottom rows show the first two W-DMD components of the raw images respectively. As can be observed, although the raw images are not visibly different from one another, the decomposed components by W-DMD clearly shows that the texture properties of both types of images are distinctive. In particular, the first W-DMD component can capture the maximum variances at column levels whereas its second component captures the residuals or noise. Therefore, our preliminary visual examination shows that W-DMD is potentially a useful approach not only to solve FV spoof detection problem, but also for addressing the biometric person authentication task; however, the latter topic falls outside the scope of this study.

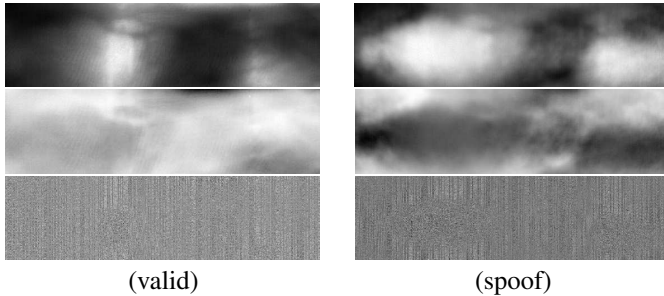


Fig. 1. Top row shows the raw images for valid access and spoof from the same person. Middle row shows the W-DMD component-1 of the raw image and bottom row shows the W-DMD component-2 of the raw image.

2) *Texture methods categorization:* Since our proposed work is based on texture analysis, we briefly compare below the characteristics of existing approaches in Table I. The table shows the different texture features commonly reported in the literature; they include Discrete Cosine Transform (DCT), Local Binary Pattern (LBP), Histogram of Gradient (HOG), Discrete Wavelet Transform (DWT) and a number of filters. We propose W-DMD as a novel addition to the above list; in doing so, we also develop the existing DMD method by applying it on a single image. Our pipeline involves: (i) decomposing candidate images into components and (ii)

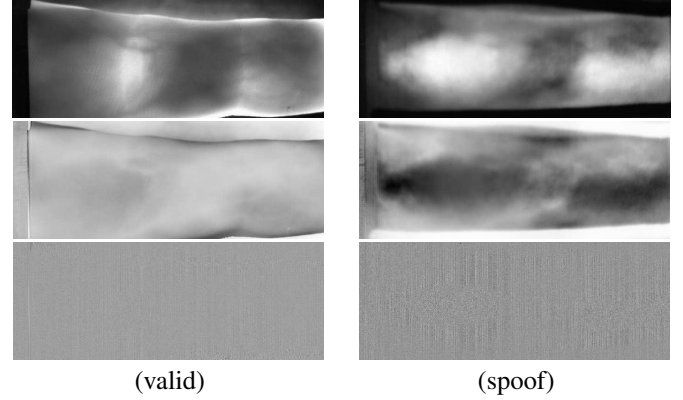


Fig. 2. Top row shows the raw images for valid access and spoof from the same person. Middle row shows W-DMD component-1 of the raw image and bottom row shows W-DMD component-2 of the raw image.

classification via Support Vector Machines with a minimum intersection kernel. The pipeline is shown in Figure 3.

TABLE I. CATEGORISATION OF THE TEXTURE ANALYSIS METHODS BASED ON SPATIAL OR FREQUENCY AND LOCAL OR GLOBAL APPROACH.

	Spatial or frequency	Local or global
LBP	Spatial	Local
W-DMD	Spatial domain	Global
DWT	Frequency domain	Global and local
DCT	Frequency domain	Global and local
HOG	Spatial	Global
Filters	Spatial	Local

In reference to Table I, we divide the features by two characteristics, namely, spatial versus frequency domain; and global versus local representation. By spatial representation, we understand that a particular feature extracts information at the pixel level, whereas by frequency domain, we imply that the feature representation captures image information using Fourier transforms or other basis functions. A feature representation is considered local if it captures information from neighbouring pixels (with respect to a reference pixel). A global feature is one that summarises the information from a given image patch. The proposed W-DMD is considered a spatial approach that represents data globally. As a result, it is not necessary to divide an image into sub-blocks; although this depends on the size of the image and the information content therein.

C. Organisation

The paper is structured as follows: Section II presents W-DMD and other texture extraction methods, including filter-based methods. Section III and Section IV present performance criteria based on EER (F-ratio) and the dataset used in this study, respectively. Section V presents our experimental protocol and results. Finally, discussions and conclusions are drawn in Section VI.

II. METHODOLOGY

As we have indicated in the introduction a number of pre-existing methods suitable for texture characterization within a spoofing-detection context, we in this section briefly present how the alternative texture based methods have been used for our evaluation along with proposed Windowed Dynamic

Mode Decomposition (W-DMD). We will also present the classification method employed.

The overall process pipeline used for evaluation is shown in Figure 3. First, an image is processed using one of the texture-based methods; output features are then fed into a trained SVM classifier in order to classify whether the processed image is a valid or spoof.

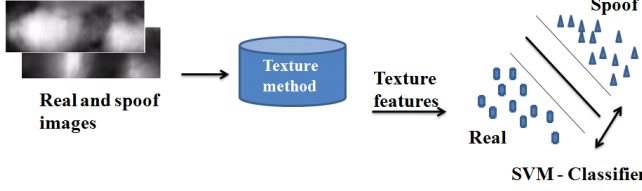


Fig. 3. Flow chart showing the steps involved in the methodological pipeline.

A. Window Dynamic Mode Decomposition (W-DMD)

Let P be an image whose columns are given by $[p_1, p_2, p_3, \dots, p_n]$. Since the consecutive columns or rows of an image are highly correlated to one another, we will assume that there exists a linear mapping A from one column to the consecutive in the sequence. This enables us to represent the image P within the span of krylov subspace [16], [2].

$$P = [p_1, Ap_1, A^2p_1, A^3p_1, \dots, A^{N-1}p_1] \quad (1)$$

In this study, we limit the krylov span for 3 columns and proceed with a sliding window approach to mimic the sequential aspect of standard DMD [2].

$$P = [p_1, Ap_1, A^2p_1]. \quad (2)$$

$$\begin{aligned} [p_2, p_3] &= A[p_1, p_2]. \\ S_2 &= AS_1. \end{aligned} \quad (3)$$

Solving eigenvectors for A is computationally more expensive. Hence, we represent Equation 3 as a standard Arnoldi iteration problem which introduces H . The eigenvalues from H approximate some of the eigenvalues of the full system A .

$$\begin{aligned} AS_1 &\approx S_1H. \\ S_2 &\approx S_1H. \\ AS_1 &\approx S_2 \approx S_1H. \end{aligned} \quad (4)$$

The associated eigenvectors of H thus provide the coefficients for the linear combination that is necessary to express the spatial variations from one column of image to another.

let us define L and U to be the matrices as a result of applying LU decomposition on S_1 and L^+ be the pseudo inverse of L . The H matrix can then be calculated by the equation:

$$H = U^{-1}L^+S_2 \quad (5)$$

For the first three columns we obtain two dynamic modes (in general for N columns, we get $N-1$ DMD modes [2]). In the next step we exclude the first column and consider columns

$\{2, 3, 4\}$, followed by $\{3, 4, 5\}$ and $\{4, 5, 6\}$ and so on as shown in Figure 4, for capturing these two dynamic modes. The first dynamic mode DM captures the intensity variations across the columns and second dynamic mode NM captures the noise residuals. Finally, we concatenate all of the DM s to obtain W-DMD component-1 (W-DMD (C1)) and similarly concatenation of the NM s produce W-DMD component-2 (W-DMD (C2)). This procedure is summarised in Algorithm 1).

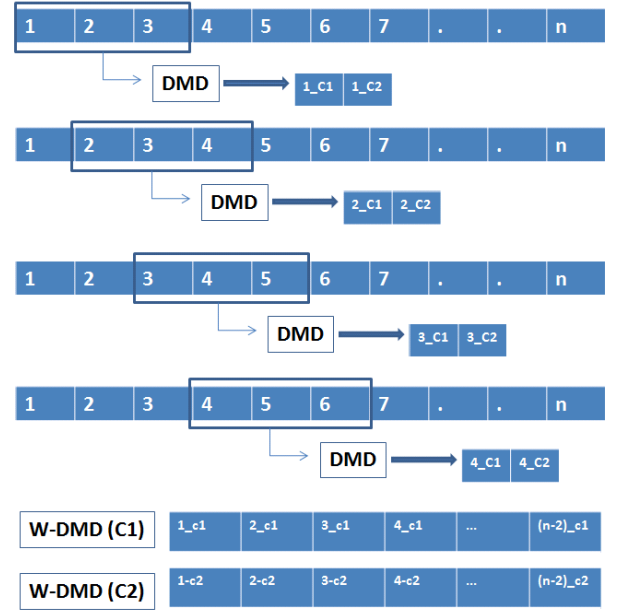


Fig. 4. Methodological pipeline showing the working mechanism of W-DMD.

Algorithm 1 W-DMD

Input: Sequences of columns in an image $P = [p_1, p_2, p_3, \dots, p_n]$.
Output: Dynamic image DM, residual image NM.
1: **for** $\{i = 1 \text{ to } n - w\} \% w = 2$
2: $S = [p_i, p_{i+1}, p_{i+2}]$
3: $[DM(i), NM(i)] = \text{get-WDMD}(S)$
4: **endfor**

Algorithm 2 get-WDMD

Input: Sequence set of columns in a window $S = [p_1, p_2, p_3]$
Output: Dynamic column DM, residual column NM.
 $S_1 \leftarrow [p_1, p_2]$
 $S_2 \leftarrow [p_2, p_3]$
 $[L \ U] \leftarrow \text{lu}(S_1)$
4: $H \leftarrow (U^{-1}L^+S_2)$
 $[Vec \ Eval] \leftarrow \text{eig}(H)$
6: $DM \leftarrow \text{abs}(S_1 Vec(:, 1))$
 $NM \leftarrow \text{abs}(S_1 Vec(:, 2))$

B. Alternative methods

Using uniform ($u2$) *Local Binary Patterns* (LBP) [12], pixels P in a finger vein image are attributed a binary code based on their relative intensities within the neighbourhood R . In this study, we have considered $P = 8$ for block-size = 3×3 and radii = 1, i.e 3-LBP_{8,1}^{u2}. *Discrete Wavelet Transform*

TABLE II. TABLE SHOWING THE SIZE OF THE RESULTANT FEATURE VECTOR FOR EACH OF THE METHODS USED IN THIS STUDY.

	Cropped	Full
LBP	1x531	1x531
DWT	1x36	1x70
DCT	1x400	1x400
HoG	1x81	1x81
Entropy	1x138	1x270
STD	1x138	1x270
Range	1x138	1x270
W-DMD	1x3330	1x6550
W-DMD+LBP	1x531	1x531
Raw	1x3390	1x6650

(DWT, Haar) [19] divide the input finger vein image into four non-overlapping multi-resolution sub-bands LL, LH, HL and HH. Sub-band LL represent the coarse-scale DWT coefficients while the sub-bands LH, HL and HH represent the fine-scale DWT coefficients. We have used the LH sub band in this study. *Discrete Cosine Transform (DCT)* [20] captures the intrinsic properties of a finger vein image within few coefficients [21]. For instance, the top $20 \times 20 = 400$ DCT coefficients are typically sufficient to represent an image and hence considered this as a feature vector in our analysis. *Histogram of Gradients (HOG)* descriptor technique counts occurrences of gradient orientation in a localised region of interest (ROI) [22], [23]. *Filter based methods* such as *Entropy*, *Standard Deviation* and *Range* have also been used with a block of size 3×3 and stored as feature values for each block.

The finger vein images in the dataset were reduced to a size of 20% of their original size prior to processing with the aforementioned methods. Table II shows the size of the resultant feature vector obtained after processing a finger vein image.

C. SVM-Minimum Intersection Kernel

In the following evaluation, the Minimum Intersection Kernel mapping is applied within an SVM context to transform the data into an appropriate high dimensional space in order to increase the likelihood of linear separability [25] [26]. This kernel having been shown to be efficient in image classifications [27] is defined:

$$K(x, y) = \sum_{i=1}^n \min(x_i, y_i) \quad (6)$$

Here, n is the number of dimensions in the feature vector and x and y are the features. We note that this kernel has no training parameter. For the SVM optimisation procedure, we set the slack parameter, to be 0.01 for all the experiments.

III. PERFORMANCE CRITERIA

Let $\mathcal{T}$ be the domain of SVM output. The decision is made by comparing the SVM output $t \in \mathcal{T}$ with a decision threshold, $\Delta \in \mathcal{T}$, as follows:

$$\text{decision}(t) = \begin{cases} \text{valid access} & \text{if } t > \Delta \\ \text{spoof/attack} & \text{otherwise,} \end{cases} \quad (7)$$

In order to calculate errors, we introduce match and non-match score sets, $\mathcal{Y}_1 \subset \mathcal{Y}$ and $\mathcal{Y}_0 \subset \mathcal{Y}$, respectively. This

can result in two errors, namely, false rejection and false acceptance, the rates of which are calculated as follows:

$$\text{FRR}(\Delta) \equiv P(t < \Delta | \omega_1) \quad (8)$$

$$\approx \frac{|y|y \in \mathcal{Y}_1, y < \Delta|}{|\mathcal{Y}_1|} \quad (9)$$

$$\text{FAR}(\Delta) \equiv 1 - P(t < \Delta | \omega_0), \quad (10)$$

$$\approx \frac{|y|y \in \mathcal{Y}_1, y < \Delta|}{|\mathcal{Y}_1|} \quad (11)$$

respectively, where the conditioning variable ω_1 indicates that the comparison is due to a positive class (valid access) whereas ω_0 indicates that the comparison is due to a negative class (spoof/attack).

Note that FRR is an increasing function of the decision threshold; whereas FAR is a decreasing function of the decision threshold. The two error rates cross over each other on Equal Error Rate, i.e.,

$$\text{EER}(\Delta_{\text{EER}}) = \text{FAR}(\Delta_{\text{EER}}) = \text{FRR}(\Delta_{\text{EER}})$$

where $\Delta_* \in \mathcal{Y}$ is the unique decision threshold of the EER. In practice, EER is found by searching for a threshold that minimises the absolute difference between FAR and FRR.

$$\Delta_{\text{EER}} = \arg \min_{\Delta} |\text{FAR}(\Delta) - \text{FRR}(\Delta)|$$

Since most of our experiments resulted in no observable classification errors, we resolve to estimating the system performance using parametric criteria such as F-ratio, d-prime statistics and other criteria [28]. These criteria assume that the classifier output conditioned upon the class label (valid or spoof finger veins) is normally distributed. Since we use SVM in all cases, the output scores are not skewed and can roughly be described by the first (mean μ) and second order (variance σ) moments, i.e., μ^1 and σ^1 for the valid samples and μ^0 and σ^0 for the spoof samples, respectively. He

A. EER based on F-ratio

A large value of F-ratio implies higher separability; and this can be measured even when no error is observed. The F-ratio is defined as:

$$\text{F-ratio} = \frac{\mu^1 - \mu^0}{\sigma^1 + \sigma^0}. \quad (12)$$

The corresponding Equal Error Rate (EER), in which one assumes that the probabilities of false acceptance and false rejection are equal, can be obtained from the F-ratio in a closed form, as follows:

$$\text{EER} = \frac{1}{2} - \frac{1}{2} \text{erf} \left(\frac{\text{F-ratio}}{\sqrt{2}} \right) \equiv \text{EER F-ratio}, \quad (13)$$

where “erf” is the error function.

$$\text{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z \exp[-t^2] dt, \quad (14)$$

In [29], it is shown that the theoretical EER found using F-ratio correlates well with the empirical EER obtained by error counting.

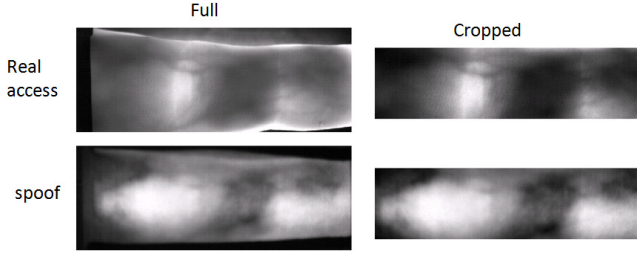


Fig. 5. Dataset divided into full and cropped finger vein images for valid and spoof samples.

IV. DATASET

IDIAP's FV-Spoofing-Attack ² dataset is considered for this study. The dataset consists of cropped (Figure 5(right)) and full versions (Figure 5(left)) of valid accesses and printed attacks from 110 clients (440 samples). Table III shows the number of samples divided into protocols.

TABLE III. NUMBER OF SAMPLES DIVIDED INTO PROTOCOLS IN THE DATASET.

Protocol	Training set	Development set	Test set
<i>full</i>	120	120	200
<i>cropped</i>	120	120	200

V. EXPERIMENTS AND RESULTS

Both valid and spoof images are collected for training, development and testing sets respectively (recall our methodological procedure from Figure 3). For each valid and spoof image the texture features are calculated and introduced to the SVM classifier with min kernel and near-boundary coefficient value = 0.01. EER based on F-ratio EER (F) and F-ratio are computed on the output of the SVM scores.

A. Selection of the W-DMD component.

W-DMD decomposes an image into two components (when $w = 3$), one capturing intensity variations in the image and the other the noise residuals. We therefore wish to establish which of these two components is important in detecting spoof images. We thus extract the W-DMD components from the train, development and test data sets, with texture features from the development set fed into the trained SVM classifier to detect whether the input image is a spoof or not in order to use the class density distributions to obtain a threshold value (point of intersection of FAR and FRR scores) from the development set. Later, this threshold value is used on the test set to discriminate between valid and spoofed videos.

TABLE IV. PERFORMANCE OF THE W-DMD COMPONENTS ON CROPPED AND FULL IMAGE DATA.

	Cropped		Full	
	EER (F)%	F-ratio	EER (F)%	F-ratio
W-DMD (C1)	1.59	2.14	0.08	3.15
W-DMD (C2)	7.81	1.41	1.80	2.09

Our findings on application of this classification procedure are that W-DMD component-1, which appears to capture the variation of light reflections within images performed well when compared to W-DMD component-2. The results on cropped images recorded an EER(F) ³ of 1.5% for W-DMD

component-1 when compared to component-2 which has given 7.8%. For the full images, the classification achieved better results compared to cropped images, presumably since these images possess finger boundaries and light reflections. Results on full images again show the strength of W-DMD component-1, recording EER(F) of 0.8% compared to 1.8% obtained by W-DMD component-2. These results are shown in Table IV.

B. Comparison with state-of-the-art methods.

To evaluate the performance of the proposed method on the FV-spoofing dataset, we compare it to the indicated state-of-the-art texture analysis methods: 1) Block - LBP, 2) DWT, 3) DCT, 4) HOG and 5) filter based methods including: a) Entropy, b) Standard deviation and c) Range.

1) *On full images:* In the case of full images, we see that design filters and DCT methods give a lower performance when compared to raw input images (which is considered as a baseline in our experiments). We see that the methods W-DMD, W-DMD+LBP, DWT, LBP and HoG have all shown better performance than the baseline. W-DMD achieves the greatest F-ratio (3.15) in comparison to the rest of the methods. These results are shown in Table V. The results in performance order are shown Figure 6.

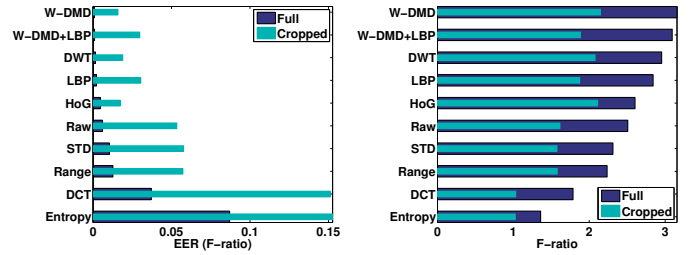


Fig. 6. Comparison of EER (F) & F-ratio on the state of the art methods for full and cropped finger vein images.

2) *On cropped images:* Similar to the results on the full image set, the design filters and DCT method give a lower performance compared to the raw image. The W-DMD method has again achieved remarkable performance with EER (F-ratio) of 1.5% and F-ratio of 2.14 when compared to rest of the methods as shown in Figure 6.

The overall performance of W-DMD with respect to the rest of the methods considered in this experiment is given in Table V.

TABLE V. PERFORMANCE OF THE STATE OF THE ART METHODS IN COMPARISON WITH THE W-DMD METHOD FOR CROPPED AND FULL IMAGE DATA.

	Cropped		Full	
	EER (F)(%)	F-ratio	EER (F)(%)	F-ratio
W-DMD	1.59	2.14	0.08	3.15
W-DMD+LBP	2.98	1.88	0.10	3.09
DWT	1.89	2.07	0.16	2.95
LBP	3.04	1.87	0.22	2.84
HoG	1.74	2.11	0.46	2.60
RAW	5.34	1.61	0.61	2.50
STD	5.77	1.57	1.04	2.31
Range	5.72	1.57	1.27	2.23
DCT	15.11	1.03	3.70	1.78
Entropy	15.26	1.02	8.70	1.35

Comparing both experimental configurations, it can be observed that the cropped images are harder to classify. This

²<https://www.idiap.ch/dataset/fvspoofingattack>

³EER calculated from F-ratio

is expected since the border information is not available. As a result, the performance of the classifier is entirely dependent on the capability of the feature to extract micro-texture information without using additional cues.

VI. DISCUSSIONS & CONCLUSIONS

This study shows the significance of the proposed novel method, W-DMD, as a viable feature extraction method when coupled with an SVM to effectively detect spoof samples. We applied W-DMD on finger vein images for valid and print attacks from 110 clients (240 (training) + 240 (development)). The results demonstrate that the proposed W-DMD is very promising in tackling the print attack challenge.

W-DMD can extract local variations as low rank representation inside an image capturing light reflections, illuminations and planar effects which are typically different for valid access and spoof images. This is the main intrinsic difference between W-DMD features and other existing features. The SVM with Minimum Intersection Kernel works particularly well with the W-DMD features, with the convenience of no additional parameter required for training, other than setting the near boundary coefficient value to 0.01. The entire pipeline proves to be efficient in comparison to the performance of other state of the art methods such as LBP, DWT, HOG and filter methods including entropy, std and range. Local-spatial texture based methods (LBP), Global-spatial (HOG) and frequency based (DWT) methods have shown better results when compared to the baseline raw input image. In future work, we would like to explore the effect of window size within the W-DMD method.

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